「Def. Let F be a field. Given a polynomial $f(x) \in F[x]$ over F, the Extension

$$E \overset{\text{def}}{=} F(\alpha_1, \cdots, \alpha_n) \quad \text{where } \alpha_1, \cdots, \alpha_n \in \bar{F} \text{ are roots of } f.$$

is called the splitting field of $f(x)$.
Further, the collection of polynomials $\{f_i(x) \mid i \in I\}$, the intersection of splitting fields of f_i is also called the splitting field.

「Thm. Let $\varphi : F \to F'$ be an isomorphism of fields.
Given $f(x) = a_n x^n + \cdots + a_1 x + a_0 \in F[x]$, let $f'(x) = \varphi(a_n) x^n + \cdots + \varphi(a_0) \in F'[x]$.
Let E be the splitting field for $f(x)$ over F,
and E' be the splitting field for $f'(x)$ over F'.
Then, φ can be extended on E to E', isomorphically.